

Q.2 Differentiate the following functions.

$$f(x) = (x^3 + 2x)e^x$$

$$g(x) = \sqrt{x} e^x$$

$$y = \frac{x}{e^x}$$

$$y = \frac{e^x}{1 - e^x}$$

$$g(x) = \frac{1 + 2x}{3 - 4x}$$

$$G(x) = \frac{x^2 - 2}{2x + 1}$$

Q.3 Find an equation of the tangent line to the curve at the given point.

$$y = \sqrt[4]{x}, \quad (1, 1)$$

$$y = x^4 + 2x^2 - x, \quad (1, 2)$$

$$y = x^4 + 2e^x, \quad (0, 2)$$

$$y = x^2 - x^4, \quad (1, 0)$$

Q.4 Find $\frac{dy}{dx}$ by implicit differentiation.

$$x^3 + y^3 = 1$$

$$2\sqrt{x} + \sqrt{y} = 3$$

$$x^2 + xy - y^2 = 4$$

$$2x^3 + x^2y - xy^3 = 2$$

Q.5 Find the derivatives following functions.

$$F(x) = (x^4 + 3x^2 - 2)^5 \quad F(x) = (4x - x^2)^{100}$$

$$F(x) = \sqrt{1 - 2x}$$

$$f(x) = \frac{1}{(1 + \sec x)^2}$$

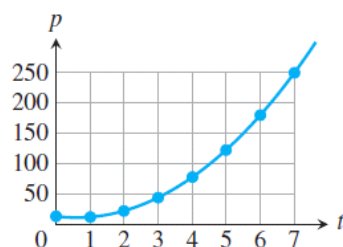
$$f(z) = \frac{1}{z^2 + 1}$$

$$f(t) = \sin(e^t) + e^{\sin t}$$

Word Problems

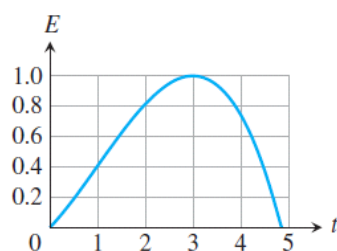
Interpreting Derivative Values

23. **Growth of yeast cells** In a controlled laboratory experiment, yeast cells are grown in an automated cell culture system that counts the number P of cells present at hourly intervals. The number after t hours is shown in the accompanying figure.



- Explain what is meant by the derivative $P'(5)$. What are its units?
- Which is larger, $P'(2)$ or $P'(3)$? Give a reason for your answer.
- The quadratic curve capturing the trend of the data points (see Section 1.4) is given by $P(t) = 6.10t^2 - 9.28t + 16.43$. Find the instantaneous rate of growth when $t = 5$ hours.

- 24. Effectiveness of a drug** On a scale from 0 to 1, the effectiveness E of a pain-killing drug t hours after entering the bloodstream is displayed in the accompanying figure.



- At what times does the effectiveness appear to be increasing? What is true about the derivative at those times?
- At what time would you estimate that the drug reaches its maximum effectiveness? What is true about the derivative at that time? What is true about the derivative as time increases in the 1 hour *before* your estimated time?

At what points do the graphs of the functions in Exercises 25 and 26 have horizontal tangents?

25. $f(x) = x^2 + 4x - 1$

26. $g(x) = x^3 - 3x$

27. Find equations of all lines having slope -1 that are tangent to the curve $y = 1/(x - 1)$.
28. Find an equation of the straight line having slope $1/4$ that is tangent to the curve $y = \sqrt{x}$.

- 97. Running machinery too fast** Suppose that a piston is moving straight up and down and that its position at time t sec is

$$s = A \cos(2\pi bt),$$

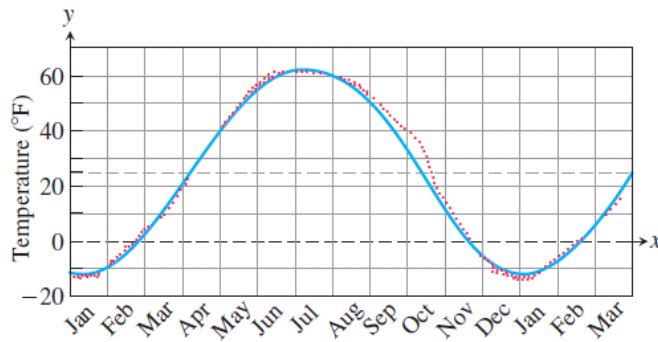
with A and b positive. The value of A is the amplitude of the motion, and b is the frequency (number of times the piston moves up and down each second). What effect does doubling the frequency have on the piston's velocity, acceleration, and jerk? (Once you find out, you will know why some machinery breaks when you run it too fast.)

- 98. Temperatures in Fairbanks, Alaska** The graph in the accompanying figure shows the average Fahrenheit temperature in Fairbanks, Alaska, during a typical 365-day year. The equation that approximates the temperature on day x is

$$y = 37 \sin \left[\frac{2\pi}{365}(x - 101) \right] + 25$$

and is graphed in the accompanying figure.

- a. On what day is the temperature increasing the fastest?
- b. About how many degrees per day is the temperature increasing when it is increasing at its fastest?



103. Temperature and the period of a pendulum For oscillations of small amplitude (short swings), we may safely model the relationship between the period T and the length L of a simple pendulum with the equation

$$T = 2\pi\sqrt{\frac{L}{g}},$$

where g is the constant acceleration of gravity at the pendulum's location. If we measure g in centimeters per second squared, we measure L in centimeters and T in seconds. If the pendulum is made of metal, its length will vary with temperature, either increasing or decreasing at a rate that is roughly proportional to L . In symbols, with u being temperature and k the proportionality constant,

$$\frac{dL}{du} = kL.$$

Assuming this to be the case, show that the rate at which the period changes with respect to temperature is $kT/2$.